Supplementary Material
How Hard Is the Attack? A Complexity-Theoretic Systematization of Machine Learning Security

Halil İbrahim Kanpak
Koç University

I. Classifying the corpus

This section reports the corpus in two layers. Tables II to IV cross-tabulate features against one another, with works in the cells by citation number. Tables V and VI give one row per work, so that every cell above is traceable to the paper behind it. Every value comes from a fixed vocabulary settled before the pass began.

The cross-tabulations are the layer that carries an argument, because in them an empty cell is a combination nobody has published. The value `none` is an answer and not a missing entry, recording that a work constrains no party at all. Of the 181 works classified so far, 32 state no resource bound on any party, and 0 fall outside the frame in the sense that deleting every resource restriction leaves the main claim unchanged.

This is a partial pass. 181 of the corpus's works are classified. The remainder are not yet placed.

TABLE I: The corpus by pipeline phase (rows) against access level (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

phase	**hard label**	**scores**	**black box**	**white box**	**weights**	**gradients**	**channel**	**none**	***unassigned***	Σ
train time	[?]		[?], [?]	[?], [?], [?], [?], [?], [?], [?], [?]		[?]	[?]	[?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?]		29
test time	[?]		[?], [?], [?], [?], [?], [?], [?], [?]	[?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?]			[?]	[?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?]		41
post hoc		[?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?]	[?], [?], [?], [?], [?], [?], [?], [?]	[?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?]	[?]	[?]		[?], [?], [?], [?], [?]	[?]	41
n/a		[?]						[?], [?]		70
Σ	2	12	18	38	1	2	2	105	1	181

TABLE II: The corpus by constrained party (rows) against resource unit (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies. A dagger marks works where the resource appears only as a side condition rather than as a parameter the work assesses on purpose.

party	**time**	**circuit size**	**queries**	**samples**	**memory**	**algo class**	**parameters**	**perturbation**	**corruption**	**rounds**	**privacy budget**	**circuit depth**	**success p**
attacker	[?], [?]†	[?], [?], [?], [?], [?], [?]	[?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?]†	[?], [?]†	[?], [?], [?], [?]	[?], [?], [?], [?], [?]		[?], [?]†	[?], [?], [?], [?], [?]			[?]	
learner	[?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?]	[?]	[?]†	[?], [?], [?], [?], [?], [?], [?], [?], [?], [?]	[?]	[?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?]	[?], [?]						
auditor	[?], [?], [?], [?], [?], [?], [?]†			[?], [?]		[?], [?], [?]							
mechanism	[?], [?]				[?], [?]	[?]†					[?], [?]†		
none													
Σ	54	7	12	14	7	22	2	23	5	0	2	1	0

TABLE III: The corpus by resource unit (rows) against tightness of the bound (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

unit	**unstated**	**poly**	**quasi-poly**	**class**	**budget**	**conditional**	**exponent**	**measured**	Σ
time	[?], [?]	[?], [?]	[?]		[?], [?]	[?], [?], [?], [?], [?], [?]	[?], [?], [?], [?], [?], [?], [?], [?]	[?], [?], [?], [?], [?], [?]	54
circuit size		[?]	[?], [?], [?]		[?]	[?]	[?]		7
queries					[?], [?], [?], [?], [?]		[?], [?], [?], [?]	[?], [?], [?]	12
samples				[?], [?]	[?], [?], [?], [?], [?]		[?], [?], [?], [?], [?], [?]	[?]	14
memory					[?]	[?]	[?], [?], [?], [?], [?]		7
algo class				[?], [?]					22
parameters					[?], [?]				2
perturbation					[?], [?]				23
corruption					[?], [?], [?], [?]			[?]	5
rounds									0
privacy budget					[?], [?]				2
circuit depth							[?]		1
success prob									0
none	[?], [?]								32
Σ	34	30	4	24	45	8	25	11	181

TABLE IV: The corpus by attack family (rows) against constrained party (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

family	**attacker**	**learner**	**auditor**	**mechanism**	**none**	Σ
membership	[?], [?], [?], [?], [?]		[?]		[?], [?], [?], [?], [?], [?]	12
attribute						0
reconstruct	[?], [?], [?], [?], [?], [?]				[?], [?], [?], [?]	10
distribution	[?], [?]					2
extract	[?], [?], [?], [?], [?]				[?]	6
invert						0
forge	[?], [?], [?], [?], [?]				[?], [?], [?]	8
evade	[?], [?]	[?], [?], [?], [?], [?], [?], [?]	[?]		[?], [?]	37
evade audit	[?], [?]		[?]		[?], [?]	5
inject			[?]			1
poison	[?], [?], [?], [?]	[?], [?], [?], [?]				8
backdoor	[?], [?]		[?], [?], [?], [?], [?]		[?], [?], [?]	10
multiple					[?], [?]	2
none	[?], [?]	[?], [?]	[?], [?], [?]	[?], [?], [?], [?], [?], [?]	[?], [?], [?], [?], [?], [?], [?], [?]	77
unassigned	[?]			[?]	[?]	3
Σ	85	45	12	7	32	181

TABLE V: Per-work reference layer: attack scenario and threat model.

#	Target	Family	Phase	Access	Write	Frame
[?]	train data	membership	post hoc	none	read only	none
[?]	train data	membership	post hoc	scores	read only	none
[?]	train data	membership	post hoc	scores	read only	none
[?]	train data	membership	train time	white box	corpus	compute
[?]	train data	membership	post hoc	weights	read only	statistical
[?]	train data	membership	post hoc	scores	read only	compute
[?]	train data	membership	post hoc	scores	read only	compute
[?]	train data	membership	post hoc	scores	read only	compute
[?]	train data	membership	post hoc	scores	read only	none
[?]	train data	membership	post hoc	white box	read only	none
[?]	train data	membership	post hoc	none	none	none
[?]	train data	membership	post hoc	white box	read only	compute
[?]	train data	reconstruct	post hoc	white box	read only	none
[?]	train data	reconstruct	post hoc	none	read only	compute
[?]	train data	reconstruct	n/a	scores	read only	compute
[?]	train data	reconstruct	post hoc	white box	read only	none
[?]	train data	reconstruct	train time	gradients	read only	compute
[?]	train data	reconstruct	post hoc	scores	read only	compute
[?]	train data	reconstruct	post hoc	white box	read only	compute
[?]	train data	reconstruct	post hoc	white box	read only	none
[?]	train data	reconstruct	post hoc	black box	read only	none
[?]	train data	reconstruct	post hoc	scores	read only	compute
[?]	train data	distribution	post hoc	scores	read only	compute
[?]	train data	distribution	train time	channel	corpus	compute
[?]	parameters	extract	post hoc	black box	read only	compute
[?]	parameters	extract	post hoc	black box	read only	none
[?]	parameters	extract	post hoc	black box	read only	compute
[?]	parameters	extract	post hoc	scores	read only	compute
[?]	parameters	extract	post hoc	black box	read only	compute
[?]	parameters	extract	post hoc	scores	read only	compute
[?]	training run	forge	post hoc	white box	corpus	none
[?]	none	forge	n/a	none	none	compute
[?]	train data	forge	post hoc	gradients	corpus	none
[?]	training run	forge	train time	white box	model	compute
[?]	training run	forge	post hoc	white box	corpus	compute
[?]	training run	forge	train time	white box	model	compute
[?]	training run	forge	post hoc	white box	model	compute
[?]	training run	forge	train time	white box	corpus	none
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	none	read only	capability
[?]	outputs	evade	test time	none	read only	compute
[?]	outputs	evade	test time	none	none	compute
[?]	outputs	evade	test time	none	read only	capability
[?]	outputs	evade	test time	none	read only	capability
[?]	outputs	evade	test time	none	read only	statistical
[?]	outputs	evade	test time	none	read only	capability
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	none	read only	capability
[?]	outputs	evade	test time	hard label	read only	compute
[?]	outputs	evade	test time	black box	read only	compute
[?]	outputs	evade	test time	none	read only	capability
[?]	outputs	evade	test time	none	read only	capability
[?]	outputs	evade	test time	none	read only	capability
[?]	outputs	evade	test time	black box	read only	compute
[?]	outputs	evade	test time	white box	read only	none
[?]	outputs	evade	test time	black box	read only	none
[?]	outputs	evade	test time	none	read only	capability
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	white box	read only	capability

Table V, continued

#	Target	Family	Phase	Access	Write	Frame
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	none	read only	statistical
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	white box	read only	statistical
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	channel	channel	capability
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	black box	read only	compute
[?]	outputs	evade	test time	black box	read only	compute
[?]	outputs	evade	test time	white box	read only	capability
[?]	outputs	evade	test time	white box	none	statistical
[?]	none	evade	train time	none	none	compute
[?]	training run	evade audit	post hoc	none	none	none
[?]	training run	evade audit	post hoc	none	responses	statistical
[?]	outputs	evade audit	test time	black box	read only	none
[?]	outputs	evade audit	test time	black box	responses	compute
[?]	training run	evade audit	train time	white box	model	compute
[?]	outputs	inject	test time	none	channel	statistical
[?]	train data	poison	train time	none	corpus	capability
[?]	train data	poison	train time	white box	corpus	capability
[?]	train data	poison	train time	white box	corpus	capability
[?]	train data	poison	train time	black box	corpus	capability
[?]	train data	poison	train time	none	corpus	compute
[?]	train data	poison	train time	none	corpus	compute
[?]	train data	poison	train time	none	corpus	compute
[?]	train data	poison	train time	none	corpus	compute
[?]	parameters	backdoor	post hoc	white box	read only	compute
[?]	parameters	backdoor	post hoc	white box	read only	none
[?]	outputs	backdoor	post hoc	black box	read only	compute
[?]	parameters	backdoor	post hoc	white box	read only	none
[?]	parameters	backdoor	post hoc	white box	model	compute
[?]	parameters	backdoor	post hoc	black box	read only	compute
[?]	parameters	backdoor	train time	black box	model	compute
[?]	train data	backdoor	train time	none	corpus	none
[?]	parameters	backdoor	train time	white box	model	compute
[?]	parameters	backdoor	train time	none	corpus	capability
[?]	none	multiple	n/a	none	none	none
[?]	none	multiple	n/a	none	none	none
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	none
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	train time	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	train data	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	train data	none	n/a	none	none	compute
[?]	training run	none	train time	hard label	read only	compute
[?]	none	none	n/a	none	none	compute
[?]	train data	none	train time	none	none	capability
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute

Table V, continued

#	Target	Family	Phase	Access	Write	Frame
[?]	none	none	n/a	none	none	none
[?]	none	none	n/a	none	none	compute
[?]	train data	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	train time	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	outputs	none	test time	black box	corpus	none
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	none
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	outputs	none	post hoc	black box	read only	compute
[?]	none	none	n/a	none	none	none
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	none
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	train time	none	none	compute
[?]	none	none	n/a	none	none	none
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	train time	none	none	compute
[?]	none	none	train time	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	train time	none	none	compute
[?]	train data	none	train time	none	none	compute
[?]	none	none	n/a	none	none	compute
[?]	none	none	n/a	none	none	none
[?]	train data	·	post hoc	white box	read only	capability
[?]	outputs	·	test time	none	responses	none
[?]	train data	·	post hoc	·	read only	statistical

TABLE VI: Per-work reference layer: adversary assumptions and claim shape. Same rows and same order as Table V.

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[?]	none	none	unstated	stated	none	none	·	empirical	stated
[?]	none	none	unstated	stated	none	none	·	empirical	stated
[?]	none	none	unstated	stated	none	none	·	empirical	stated
[?]	attacker	time	poly	stated	none	none	explicit	conditional	stated
[?]	attacker	samples	exponent	stated	samples	·	explicit	universal	stated
[?]	attacker	queries	measured	stated	none	none	implicit	empirical	stated
[?]	auditor	time	unstated	stated	none	none	implicit	conditional	stated
[?]	attacker	time	measured	stated	none	none	implicit	empirical	stated
[?]	none	none	unstated	stated	none	none	·	empirical	stated
[?]	none	none	unstated	stated	none	none	·	conditional	stated
[?]	none	none	unstated	stated	none	none	·	empirical	stated
[?]	attacker	time	poly	stated	none	queries	explicit	conditional	stated
[?]	none	none	unstated	stated	none	·	·	universal	stated
[?]	attacker	time	poly	stated	time	none	explicit	conditional	stated
[?]	attacker	time	poly	absent	perturbation	queries	explicit	universal	stated
[?]	none	none	unstated	stated	none	·	·	existential	stated
[?]	attacker	time	measured	absent	none	none	implicit	empirical	stated
[?]	attacker	queries	exponent	stated	samples	none	explicit	conditional	stated
[?]	attacker	queries	budget	absent	privacy budget	none	explicit	universal	stated
[?]	none	none	unstated	stated	none	none	·	conditional	stated
[?]	none	none	unstated	n/a	none	none	·	empirical	stated
[?]	attacker	time	poly	stated	none	queries	explicit	conditional	stated
[?]	attacker	time	poly	stated	·	none	explicit	conditional	stated
[?]	attacker	time	poly	stated	time	none	explicit	conditional	stated
[?]	attacker	queries	budget	absent	none	none	explicit	conditional	stated
[?]	none	none	unstated	absent	none	none	·	empirical	stated
[?]	attacker	time	measured	stated	none	none	explicit	empirical	stated
[?]	attacker	queries	measured	absent	none	none	implicit	empirical	stated
[?]	attacker	algo class	class	absent	none	none	explicit	existential	stated
[?]	attacker	queries	measured	stated	none	time	implicit	empirical	stated
[?]	none	none	unstated	n/a	none	none	·	existential	stated
[?]	attacker	memory	exponent	stated	memory	none	explicit	universal	stated
[?]	none	none	unstated	absent	none	none	·	empirical	stated
[?]	attacker	time	budget	n/a	time	none	implicit	conditional	stated
[?]	attacker	time	measured	stated	none	none	implicit	empirical	stated
[?]	attacker	time	measured	stated	none	none	explicit	empirical	stated
[?]	attacker	time	unstated	absent	none	none	explicit	conditional	stated
[?]	none	none	unstated	n/a	none	none	·	empirical	stated
[?]	attacker	perturbation	budget	stated	none	none	explicit	existential	none
[?]	attacker	perturbation	budget	n/a	none	none	implicit	empirical	none
[?]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[?]	learner	algo class	class	n/a	perturbation	·	explicit	existential	stated
[?]	learner	time	poly	stated	perturbation	none	explicit	conditional	stated
[?]	attacker	perturbation	budget	n/a	none	none	explicit	existential	stated
[?]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[?]	learner	samples	budget	n/a	none	·	explicit	universal	stated
[?]	attacker	perturbation	budget	n/a	none	·	explicit	universal	stated
[?]	attacker	perturbation	budget	n/a	none	none	·	universal	stated
[?]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[?]	attacker	time	poly	n/a	none	none	·	universal	stated
[?]	attacker	circuit size	poly	n/a	none	none	·	existential	stated
[?]	attacker	perturbation	budget	n/a	none	none	explicit	conditional	stated
[?]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[?]	attacker	perturbation	budget	n/a	none	none	explicit	empirical	stated
[?]	auditor	time	exponent	stated	none	·	implicit	existential	stated
[?]	none	none	unstated	stated	none	·	·	conditional	stated
[?]	none	none	unstated	n/a	none	none	·	empirical	stated
[?]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[?]	attacker	perturbation	budget	n/a	none	none	explicit	conditional	stated

Table VI, continued

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[?]	attacker	perturbation	budget	stated	none	none	explicit	existential	none
[?]	attacker	perturbation	budget	n/a	none	none	implicit	empirical	none
[?]	learner	samples	budget	n/a	none	none	explicit	universal	stated
[?]	attacker	perturbation	budget	n/a	none	none	·	universal	stated
[?]	attacker	perturbation	budget	n/a	none	none	·	empirical	stated
[?]	attacker	perturbation	budget	n/a	samples	none	explicit	existential	stated
[?]	learner	samples	budget	stated	none	none	explicit	universal	stated
[?]	attacker	perturbation	budget	stated	none	time	explicit	empirical	stated
[?]	attacker	perturbation	budget	stated	none	none	explicit	existential	stated
[?]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[?]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[?]	attacker	queries	budget	stated	none	algo class	implicit	existential	none
[?]	attacker	time	measured	n/a	none	none	implicit	empirical	none
[?]	attacker	perturbation	budget	n/a	samples	none	explicit	universal	stated
[?]	learner	samples	budget	n/a	none	none	explicit	universal	stated
[?]	learner	time	poly	n/a	perturbation	none	explicit	conditional	stated
[?]	none	none	unstated	n/a	none	·	·	conditional	stated
[?]	auditor	samples	exponent	n/a	none	none	·	conditional	stated
[?]	none	none	unstated	n/a	none	none	·	existential	stated
[?]	attacker	queries	exponent	stated	none	time	explicit	existential	stated
[?]	attacker	time	exponent	n/a	none	·	implicit	existential	stated
[?]	auditor	samples	budget	n/a	none	none	explicit	conditional	stated
[?]	attacker	corruption	budget	stated	algo class	none	explicit	conditional	stated
[?]	attacker	corruption	budget	n/a	none	none	explicit	conditional	stated
[?]	attacker	corruption	budget	stated	none	·	explicit	conditional	stated
[?]	attacker	corruption	budget	stated	none	perturbation	explicit	universal	stated
[?]	learner	time	poly	n/a	corruption	·	explicit	existential	stated
[?]	learner	samples	class	stated	none	algo class	explicit	conditional	stated
[?]	learner	samples	class	stated	none	algo class	explicit	universal	stated
[?]	learner	queries	budget	stated	corruption	time	implicit	universal	stated
[?]	auditor	time	poly	stated	corruption	none	explicit	conditional	stated
[?]	none	none	unstated	n/a	none	·	·	existential	none
[?]	attacker	queries	budget	n/a	none	time	implicit	empirical	stated
[?]	none	none	unstated	n/a	none	none	·	existential	none
[?]	auditor	time	poly	stated	perturbation	none	explicit	existential	stated
[?]	auditor	time	poly	absent	corruption	samples	explicit	conditional	stated
[?]	auditor	time	poly	stated	none	none	explicit	existential	stated
[?]	none	none	unstated	n/a	none	none	·	empirical	none
[?]	auditor	time	poly	n/a	none	none	explicit	empirical	stated
[?]	attacker	corruption	measured	stated	none	·	explicit	empirical	stated
[?]	none	none	unstated	n/a	none	none	·	empirical	stated
[?]	none	none	unstated	n/a	none	none	·	empirical	stated
[?]	mechanism	memory	exponent	stated	none	·	explicit	existential	stated
[?]	none	none	unstated	n/a	none	none	·	conditional	stated
[?]	attacker	time	exponent	n/a	time	none	explicit	conditional	stated
[?]	learner	time	exponent	n/a	none	none	·	conditional	stated
[?]	learner	algo class	class	n/a	none	·	explicit	conditional	stated
[?]	auditor	algo class	class	n/a	none	none	explicit	universal	stated
[?]	attacker	circuit size	budget	n/a	circuit size	algo class	explicit	existential	stated
[?]	attacker	memory	exponent	n/a	memory	queries	explicit	universal	stated
[?]	attacker	algo class	class	n/a	none	algo class	explicit	conditional	stated
[?]	learner	time	conditional	n/a	none	algo class	explicit	conditional	stated
[?]	attacker	memory	exponent	n/a	memory	time	explicit	conditional	stated
[?]	attacker	algo class	class	stated	none	·	explicit	existential	stated
[?]	attacker	time	poly	n/a	none	none	·	conditional	stated
[?]	attacker	time	poly	n/a	none	none	·	conditional	stated
[?]	attacker	time	poly	stated	none	none	explicit	existential	stated
[?]	attacker	time	poly	stated	none	·	explicit	conditional	stated
[?]	attacker	circuit size	quasi-poly	stated	none	none	·	conditional	stated
[?]	mechanism	privacy budget	budget	n/a	none	none	·	empirical	stated
[?]	learner	time	poly	n/a	none	·	explicit	conditional	stated

Table VI, continued

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[?]	learner	time	conditional	stated	none	algo class	explicit	conditional	stated
[?]	none	none	unstated	n/a	none	none	·	empirical	none
[?]	mechanism	memory	conditional	stated	time	none	explicit	conditional	stated
[?]	mechanism	time	budget	stated	none	·	explicit	universal	stated
[?]	learner	algo class	class	stated	none	queries	explicit	universal	stated
[?]	attacker	algo class	class	stated	algo class	none	explicit	conditional	stated
[?]	learner	time	conditional	n/a	none	·	explicit	conditional	stated
[?]	learner	time	conditional	n/a	none	none	explicit	conditional	stated
[?]	attacker	queries	exponent	stated	queries	none	explicit	existential	stated
[?]	attacker	time	exponent	stated	time	none	explicit	conditional	stated
[?]	attacker	circuit size	exponent	n/a	none	none	explicit	existential	stated
[?]	attacker	time	conditional	n/a	none	none	explicit	conditional	stated
[?]	none	none	unstated	n/a	none	none	·	empirical	stated
[?]	learner	circuit size	conditional	n/a	none	none	explicit	universal	stated
[?]	learner	algo class	class	n/a	none	·	explicit	universal	stated
[?]	learner	samples	exponent	stated	none	algo class	explicit	conditional	stated
[?]	learner	samples	exponent	absent	none	algo class	explicit	conditional	stated
[?]	learner	time	poly	stated	none	·	explicit	conditional	stated
[?]	learner	time	poly	stated	perturbation	·	explicit	conditional	stated
[?]	none	none	unstated	n/a	none	·	·	universal	stated
[?]	learner	time	poly	n/a	none	algo class	explicit	existential	stated
[?]	mechanism	algo class	class	stated	none	time	implicit	conditional	stated
[?]	learner	algo class	class	n/a	none	samples	explicit	universal	stated
[?]	auditor	algo class	class	n/a	privacy budget	samples	explicit	existential	stated
[?]	none	none	unstated	n/a	none	none	·	conditional	stated
[?]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[?]	none	none	unstated	n/a	none	none	·	universal	stated
[?]	learner	algo class	class	n/a	none	time	explicit	conditional	stated
[?]	learner	algo class	class	n/a	none	queries	explicit	universal	stated
[?]	learner	time	poly	n/a	none	none	explicit	conditional	stated
[?]	attacker	time	poly	n/a	privacy budget	none	explicit	conditional	stated
[?]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[?]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[?]	learner	algo class	class	n/a	none	·	explicit	conditional	stated
[?]	none	none	unstated	n/a	none	none	·	conditional	stated
[?]	learner	algo class	class	n/a	none	samples	explicit	existential	stated
[?]	attacker	queries	exponent	n/a	queries	none	explicit	universal	stated
[?]	attacker	algo class	class	stated	algo class	none	explicit	conditional	stated
[?]	learner	samples	exponent	stated	none	algo class	explicit	conditional	stated
[?]	learner	parameters	budget	stated	none	time	explicit	existential	stated
[?]	attacker	time	exponent	n/a	time	none	explicit	conditional	stated
[?]	attacker	time	exponent	n/a	none	none	explicit	conditional	stated
[?]	attacker	circuit size	quasi-poly	stated	none	none	·	conditional	stated
[?]	attacker	circuit size	quasi-poly	stated	none	none	·	conditional	stated
[?]	learner	time	exponent	n/a	none	success prob	explicit	conditional	stated
[?]	learner	time	quasi-poly	n/a	none	algo class	explicit	existential	stated
[?]	learner	memory	budget	n/a	none	none	explicit	universal	stated
[?]	learner	time	conditional	stated	corruption	algo class	explicit	conditional	stated
[?]	learner	samples	exponent	stated	none	time	explicit	universal	stated
[?]	auditor	algo class	class	n/a	none	·	explicit	conditional	stated
[?]	mechanism	time	poly	n/a	time	none	explicit	conditional	stated
[?]	attacker	circuit depth	exponent	n/a	none	queries	explicit	existential	stated
[?]	attacker	time	poly	n/a	privacy budget	none	explicit	universal	stated
[?]	attacker	memory	exponent	stated	memory	·	explicit	universal	stated
[?]	learner	parameters	budget	n/a	none	none	explicit	conditional	stated
[?]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[?]	learner	time	poly	stated	none	samples	explicit	existential	stated
[?]	none	none	unstated	n/a	none	none	·	empirical	stated
[?]	mechanism	privacy budget	budget	stated	none	none	implicit	universal	stated
[?]	none	none	unstated	n/a	none	none	·	empirical	stated

[**?**]	attacker	samples	measured	stated	none	none	implicit	empirical	stated